

T02 VERIFIED THERMAL COMFORT | O (MAX : 3 PT)

Intent : Enhance thermal comfort and promote human productivity, by ensuring that a substantial majority of building users (above 80%) perceive their environment as thermally acceptable

Summary : This WELL feature requires projects to provide high levels of thermal comfort, by determining occupant satisfaction through a survey.

Issue : Due to the strong influence of the thermal comfort, standards establish that building's thermal environment must satisfy a minimum of 80% of occupants to be considered acceptable.¹ However, despite having established standards, studies have also shown that only 11% of office buildings surveyed in the U.S. provide thermal environments that meet generally accepted goals of human satisfaction². Similarly, as many as 41% of office workers have expressed dissatisfaction with the thermal environment.^{2,3} Building users who are satisfied with their thermal environment have been shown to be more productive in the workplace,² while thermal discomfort is associated with sick building syndrome symptoms and other conditions that lead to a decrease in productivity.^{2,4}

Solutions : Building occupants are an invaluable source of information that can be used for improving the performance of buildings. Thermal comfort surveys allow projects to objectively gauge which building services and design features are or are not performing well and help to prioritize the steps needed to improve occupant thermal comfort satisfaction and workplace productivity. If survey results indicate that percentage of occupants dissatisfied with thermal conditions in the building exceeds the targeted thresholds, it is necessary to develop a detailed plan for action and commitment to address occupant dissatisfaction with thermal conditions.

PART 1 SURVEY FOR THERMAL COMFORT (MAX : 3 PT)

For All Spaces:

A post-occupancy survey is administered at least twice a year, including once in June, July or August and once in December, January or February. For buildings located in tropical regions, the survey may be administered twice a year, at least four months apart. Survey is administered at least six months after occupancy, which satisfies the following conditions:

- a. All regular occupants are invited to participate in the anonymous survey, and responses are collected from the following number of respondents:¹

Number of Regular Occupants	Minimum Number of Responses
More than 45	35% of those regular occupants
20 to 45	15 regular occupants
Less than 20	80% of those regular occupants

- b. The survey includes an assessment of overall satisfaction with thermal performance and identification of thermal comfort-related issues in accordance with either:
 1. The sample survey in Appendix T1.
 2. Any pre-approved survey listed in Part 1: Administer Project Survey in Feature C04: Occupant Survey.
- c. The results of the survey responses comply with one of the target satisfaction thresholds as specified in the table below:

Tier	Thermal Comfort Satisfaction Thresholds	Points
1	80% of regular occupants	2
2	90% of regular occupants	3

REFERENCES

1. Kim J, Schiavon S, Brager G. Personal comfort models – A new paradigm in thermal comfort for occupant-centric environmental control. *Build Environ.* 2018;132:114-124. doi:10.1016/j.buildenv.2018.01.023
2. Huizenga C, Abbaszadeh S, Zagreus L, Arens EA. Air Quality and Thermal Comfort in Office Buildings: Results of a Large Indoor Environmental Quality Survey. In: *Proceedings of Healthy Buildings. Vol III.* ; 2006:393-397. <http://escholarship.org/uc/item/7897g2f8;jsessionid=CEA1E13173D8003D5F74BD638E71785C>.
3. Frontczak M, Schiavon S, Goins J, Arens E, Zhang H, Wargocki P. Quantitative relationships between occupant satisfaction and satisfaction aspects of indoor environmental quality and building design. *Indoor Air.* 2012;22(2):119-131. doi:10.1111/j.1600-0668.2011.00745.x
4. Wargocki P. The effects of outdoor air supply rate in an office on perceived air quality, sick building syndrome (SBS) symptoms and productivity. *Indoor Air.* 2000;10(4):222-236. doi:10.1034/j.1600-0668.2000.010004222.x
5. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE 55-2013: Thermal Environmental Conditions for Human Occupancy. 2013. https://www.techstreet.com/ashrae/standards/ashrae-55-2013?product_id=1868610.